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Basis and Dimension Graded Assignment
Graded Assignment • 20 min

Due Nov 13, 11:59 PM IST

Your grade: 80%

Your latest: **80%** • Your highest: **80%** • To pass you need at least 80%. We keep your highest score.

1. Determine the dimension of the subspace S of $\mathbb{R}^{3\times 3}$ consisting of all matrices whose trace is 0. 1 / 1 point

8

Nice work

2. Determine the dimension of the subspace S of $\mathbb{R}^{3\times 3}$ consisting of all symmetric matrices. 1 / 1 point

6

Nice work

3. Determine if the given set is a basis for the vector space $\mathbb{R}^{\infty}$. 1 / 1 point

$$V = \{(1,1,1,\dots),(0,1,1,1,\dots),(0,0,1,1,1,\dots),\dots\}$$

- The set is a basis for $\mathbb{R}^{\infty}$.
- The set is not a basis for $\mathbb{R}^{\infty}$.

Nice work

4. Determine the dimension of the subspace S of $\mathbb{R}^{3\times 3}$ consisting of all matrices A such that 1 / 1 point

$$A\begin{bmatrix}1\\1\\1\end{bmatrix} = \mathbf{0}$$

6

Nice work

5. **True** or **False**: If S_1 and S_2 are infinite-dimensional subspaces of a vector space V, then $S_1 \cap S_2$ is an infinite-dimensional subspace of V. 1 / 1 point

- True
- False

Nice work

6. **True** or **False**: If every infinite set S in a vector space V is linearly dependent, then $\dim(V) = \infty$. 1 / 1 point

- True
- False

Nice work

7. **True** or **False**: If S_1 and S_2 are infinite-dimensional subspaces of a vector space V, then $S_1 = S_2$. 1 / 1 point

- True
- False

Nice work

8. Find a basis for the subspace S of $\mathbf{P}^2$ consisting of all polynomials p satisfying $p(0) = 0$. 0 / 1 point

2

Try again

9. Determine the dimension of the subspace S of $\mathbb{R}^{\infty}$ consisting of all sequences $(x_1, x_2, \dots)$ such that the series $\sum_{n=1}^{\infty} x_n$ is convergent and satisfies $\sum_{n=1}^{\infty} x_n = 0$. 0 / 1 point

Inf

Try again

10. Determine if the given set is a basis for the vector space $\mathbb{R}^{2\times 2}$. 1 / 1 point

$$\left\{\begin{bmatrix}1&1\\0&0\end{bmatrix},\begin{bmatrix}0&1\\0&1\end{bmatrix},\begin{bmatrix}0&0\\1&1\end{bmatrix},\begin{bmatrix}1&0\\1&0\end{bmatrix}\right\}$$

- The set is a basis for $\mathbb{R}^{2\times 2}$.
- The set is not a basis for $\mathbb{R}^{2\times 2}$.

Nice work